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Influence of Consumers' Temporary Affect on Ad Engagement: A Computational Research Approach

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ABSTRACT

This study examined the influence of consumers' temporary affective states during ad exposure on their engagement with different types of ads that are categorized based on theoretically derived attention-grabbing characteristics. A computational research approach was used, cross-analyzing proxy measures of real-time affective fluctuation of viewers during the 2019 Super Bowl broadcast and their tweets regarding the ads aired during the Super Bowl. The results demonstrated significant impact of consumers' temporary affective states, induced by the performance of the team they cheer for, on their engagement with different types of ads, even when they were exposed to the same set of ads during commercial breaks. Specifically, consumers in the positive affective state showed greater tendency to be drawn to engage with high semantic-affinity ads than those in the negative affective state. Consumers in the negative affective state showed greater tendency to be drawn to engage with more positively valenced ads than those in the positive affective state. This study provides theoretical contributions regarding the role of consumers' affect in their engagement with ads and practical implications for ad targeting and ad placement strategies based on consumers' temporary affect.

In the highly cluttered media environment where overabundant ads are causing consumers to experience ad fatigue, ad avoidance has become a major concern for advertisers, and winning consumer attention and engagement by cutting through the clutter has become an ever more difficult challenge. Thus, understanding what might affect and enhance consumers' ad attention and engagement upon ad exposure has become more critical than ever for advertising researchers and practitioners. In light of this need, a great deal of research has been dedicated to elucidate the influencing factors of consumers' attention to and engagement with ads.

Prior research has identified three types of factors influencing consumers' attention to ads: (1) ad content factors; (2) context factors; and (3) consumer factors. Most of the extant research falls into the first group and has identified ad personalization (e.g., Bang and Wojdyski 2016) and various attention-eliciting

message design and structural features (e.g., novel stimuli, music, loud noises, sudden movement) as significant influences on attention to ads (e.g., Lang 2000; Pieters and Wedel 2004). Some studies have also suggested that ad content inducing negative affect (e.g., fear or anxiety) can increase consumers' further involvement and attention to ads (Bradley, Angelini, and Lee 2007; Lang 1990).

The second group of research examining the context factors is growing especially in connection to the emerging technologies and has identified ad-context congruity (e.g., Moore, Stammerjohan, and Coulter 2005), media multitasking (e.g., Guitart, Hervet, and Hildebrand 2019), human density (Wilson and Suh 2018), and location congruity (Ketelaar et al. 2017) as significant influences on attention to ads. Research on the consumer factors affecting attention to ads has focused almost exclusively on the cognitive factor (e.g., brand familiarity, perceived relevance, involvement, motivation) and

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suggested that brand familiarity and perceived relevance tends to increase attention to ads, while the influences of media context involvement and motivation on ad attention are mixed (e.g., Moorman et al. 2012; Wilson, Baack, and Till 2015). Research examining the influence of consumers' affective factors on ad attention and engagement has been limited and, to the best of our knowledge, all such research has focused on the influence of consumers' affective reactions to ads (i.e., ad-induced affective responses). The influence of consumers' affective states at the moment of ad exposure on their attention to ads and engagement has received little research attention, leaving a significant gap in the literature on affect and advertising effects.

Consumers' affect ebbs and flows over the course of a day and can change depending on the media content being consumed (e.g., feeling sad while watching a sad movie), and thus consumers are likely to be exposed to ads in the presence of some types of affect, which would likely influence their reactions to the ads. Examining this possibility and addressing the important gap in the research literature, the objective of this study is to investigate the influence of consumers' temporary affective states on their selective attention to and engagement with different types of ads that are categorized based on theoretically derived attention-grabbing characteristics. To guide this study and develop hypotheses, mood management theory is applied as the main theoretical framework.

This study is situated at the intersection of research on the ad context factor and the consumer factor influencing ad attention and engagement, and presents a novel extension to the growing research literature on the role of affect in advertising effects and effectiveness. Furthermore, this study's findings could contribute to advancing theory of context-based ad targeting and developing a new computational advertising model based on it. With the development of machine-learning algorithms and context-aware recommender systems, advertisers are increasingly interested in harnessing available data indicating consumers' current contexts for precise targeting and personalization (Araujo et al. 2020; Huh and Malthouse 2020). For example, in the mobile marketing context, locational and temporal targeting adapting to consumers' current geolocations and times have been proven effective in terms of product sales (e.g., Luo et al. 2014). In the recommendation system domain, consumers' current feelings, such as their mood while listening to music (Han et al. 2010) or their emotions after seeing a certain movie (Odić et al. 2013), are also considered a type of contextual factors used in

context-aware recommendation systems in the computer science field. The current study is connected to this emerging computer science research and will make meaningful contributions.

Methodologically, this study takes an innovative computational research approach, by developing unique proxy measures indicating real-time affect fluctuation of TV viewers and their relative attentional engagement reactions to different types of ads aired during distinctive affective states. Addressing the traditional experimental methods' limitations in manipulation and measurement of affect and ad engagement, the current study method harnessing the availability and expansion of social media data is opening new possibilities that allow researchers to examine consumers' fluctuating affective states and their influence on real-time consumer responses to ads in a natural setting. This methodological approach will be discussed in detail later; first, the following section reviews research on affect and its influence on ad reactions from the psychology and communication fields and discusses mood management theory, leading to this study's hypotheses.

Literature Review and Hypotheses

Research on the Influence of Affect on Ad Reactions

Affect, often used as an umbrella term encompassing all sorts of subjective feelings, including emotions, moods, sentiments, and emotional traits (Frijda 1994), has been conceptualized in two major approaches: dimensional and discrete (Dillard and Seo 2012). According to the dimensional views that guide the majority of the extant research, affect is conceptualized and characterized in terms of (1) valence only and (2) valence plus arousal (Dillard and Seo 2012). The discrete affect models deem affect as a set of basic and qualitatively distinct emotions that capture all subjective emotional experiences.

Another relevant concept, often subsumed under the generic concept of affect, is mood. Generally defined as a subjectively perceived affective state that is pervasive, general, nonspecific, and often of low intensity, moods impart generic valence information (Schwarz and Clore 1983). Differing from emotions, which are often of high intensity, have an identifiable referent, and are associated with corrective behavioral goals, moods lack motivational specificity (Larsen 2000) and are characterized by "the absence of impulsion toward particular courses of action" (Knobloch and Zillmann 2002, p. 357). Though not involving a

referent, moods may stem from a specific emotional event concerning a specific object (Frijda 1994).

In the research on affect in the communication field, it has been examined as a precursor to subsequent processing, evaluation, and behavior in response to media stimuli, and a mechanism through which various persuasion effects occur (Nabi 2009). In this research, an important distinction is made between message-induced affect (i.e., affective states evoked by a message, most often strategically planned by the communicator to influence the recipient) and message-irrelevant, incidental affect (i.e., affective states evoked by incidents that are irrelevant to the message) (Dillard and Meijnders 2002). Of particular interest to the current study is the influence of message-irrelevant affect—namely, consumers' preexisting affective states right at the point of ad exposure, not message-induced affect that is often intended by advertising planning.

Among different types of consumer reactions to ads, engagement, often used as a blanket term referring to all kinds of committed reactions to a particular object (Kim et al. 2017), has gained increasingly significant research attention from multiple fields such as psychology, marketing, advertising and human-computer interaction (HCI). While the engagement subjects studied in the literature include consumers, customers, and audience or media users, typical engagement objects are digital media content and systems, brands, organizations, and so on (e.g., Brodie et al. 2013; O'Brien and Toms 2008).

Given the multidisciplinary nature of the engagement concept, its conceptualizations and operationalizations vary across different disciplines. However, it is generally viewed as a multidimensional concept encompassing both psychological and behavioral dimensions (e.g., Calder, Malthouse, and Schaedel 2009; O'Brien and Toms 2008). The psychological dimension generally refers to individuals' attentive, cognitive, or affective reactions to an engagement object, while the behavioral dimension denotes manifest behaviors as the engagement subjects interact with engagement objects, such as digital outreach (i.e., liking, commenting, or sharing) and behavioral avoidance (i.e., zapping behaviors) (e.g., Oh, Bellur, and Sundar 2018; Teixeira, Wedel, and Pieters 2012). For example, Teixeira and colleagues (2012) operationalized consumers' video ad engagement as attention, emotional responses, and behavioral avoidance during exposure to video ads. Arapakis et al. (2014) defined audience engagement with news as the audience's focused attention to and affective reactions to news, as

well as gaze behaviors. Depending on the research contexts and purpose, different studies have examined and measured different aspects of engagement.

The current study adopts the conceptualization of ad engagement as an expression of consumers' attention to ads manifested via their engagement behaviors on social media (e.g., commenting about specific ads during TV watching) and focuses on both the psychological and behavioral dimensions of engagement. The focus of this research is on identifying the influence of consumers' temporary affect on their attentional engagement with ads, as exhibited by their commenting behaviors on social media.

While research on the influence of consumers' preexisting affect on reactions to ads is underdeveloped in the advertising field, general psychology research tackling relevant questions offers useful insights for this study's hypotheses development. Extant research on the influence of preexisting affect converged on one idea: affect, both discrete emotions and moods, can be characterized by its valence, reflecting the hedonic tone of the affective experience (Nabi 2009). Thus, the majority of extant research on the influence of message-irrelevant affective states followed the dimensional view and examined the effects of the bipolar valence and/or arousal of affective states on information processing, evaluation, and behaviors (e.g., Di Muro and Murray 2012). It is worthwhile to note that, despite some conceptual distinctions among moods, emotions, sentiments, and feelings, these concepts have been used interchangeably with *affect* or *affective states* in the vast majority of extant research on the influence of affect in the psychology and consumer behavior fields.

Despite the growing scholarly attention to the influence of affect on attentional reactions to stimuli, most of the empirical research has been typically focused on its influence on downstream cognitive effects, such as judgment (e.g., Schwarz and Clore 1983), memory (e.g., Bower 1981), decision making (e.g., Peters et al. 2006), and behavior (for a review, see Bower and Forgas 2000). Thus, early-stage effects of affective states, particularly on attention, have been underexplored. However, studies examining the connection between affect and selective attention started with testing the mood-cognition frameworks hypothesis (e.g., Bower 1981), which postulates that mood should direct attentional processing toward mood-congruent messages. For example, Tamir and Robinson (2007) found that positive moods were significantly associated with attention to reward words,

such as *victory* and *fun*, while neutral and negative moods were unrelated.

Compared to the influence of positive affect, the association between temporary negative affect and subsequent mood-congruent attentional processing has seen limited supporting evidence. While one study found supporting evidence of a positive link between induced sad mood and greater tendency to attend to affect-congruent information (Bradley, Mogg, and Lee 1997), other studies failed to find this pattern (e.g., Chepenik, Cornew, and Farah 2007; McCabe, Gotlib, and Martin 2000). One study even found that individuals induced into stress were faster at shifting attention away from negative words compared with positive and neutral words (Ellenbogen et al. 2002).

Attempting to address the prior inconsistent findings regarding the relationship between temporary negative affect and mood-congruent attentional processing, later research examined the possibility of selective attention resulting from the mood regulatory process and found supporting evidence. One study found that, when the participants were in a bad mood, they tended to gaze at positive faces and away from negative ones (Isaacowitz et al. 2008). Another study confirmed the finding that both those participants induced into positive affect and those induced into negative affect paid more attention to happy faces (Sanchez et al. 2014). A more recent study also found that those induced into sad mood tended to take longer time to disengage attention from happy faces (Demeyer, Sanchez, and De Raedt 2017). These results lend further evidence to the notion that individuals tend to control their attentional processes to manage their affective states.

In sum, findings from the psychology research regarding the effects of incidental affect on attentional reactions to stimuli suggest the following: Positive affect is significantly related to selective attention to rewarding information, while negative affect tends to be associated with mood-incongruent attentional processing; and such effects would likely result from the mood regulatory process—motivation to alleviate the negative affect. Informed by prior literature, this study seeks to extend the research on the influence of affect on attentional reactions in a novel manner by examining the influence of temporary affective states on selective attentional engagement with different types of ads. In the context of advertising, it is plausible that motivation to manage affect could make individuals drawn to certain types of ads that would help improve or maintain positive affect and avoid ads that might induce or exacerbate negative affect. This

possibility can be explained by mood management theory (MMT), which serves as this study's main theoretical framework.

Mood Management Theory Explaining the Influence of Affect

The main proposition of MMT (Zillmann 1988a, 2000) is that individuals tend to select and engage with media content that will optimize their mood—improve/maintain positive moods and minimize/repair negative moods. Furthermore, MMT posits that this mood optimization process does not necessitate deliberate elaboration, which means that individuals may not be cognizant of their mood management motivation (Zillmann 1988a).

Regarding the characteristics of media content that might be sought out or avoided by individuals for the purpose of mood management, MMT proposes four mood-relevant characteristics of media message types—excitatory potential, absorption potential, semantic affinity, and hedonic valence—that determine their mood management potential (Zillmann 1988a). Excitatory potential of media messages refers to the arousing or stimulating level of media content. Media content that is likely to increase and heighten arousal is considered to have high excitatory potential and vice versa. Zillmann (1988a) proposed that, for purposes of reaching excitatory homeostasis, individuals in high arousal states that are construed as aversive would likely be drawn to soothing (versus arousing) content, whereas those in low arousal states would be drawn to more exciting content.

Absorption potential refers to media content's capability to interfere with consumers' preexisting mood, namely, the extent of disruption of the continual rehearsal of previous negative incidents or gratifying aspects of previous positive incidents (Zillmann 1988a). The higher the absorption potential, the stronger the interruption of the mood-perpetuating rumination on thoughts related to previous negative or positive incidents. MMT predicts that individuals in negative moods will seek to disrupt this affective state by exposing themselves to media content with highly absorbing and engaging characteristics, such as mystery movies with intriguing plots. In contrast, individuals in positive moods would be drawn to minimally absorbing content.

The third dimension of semantic affinity (Zillmann 1988b) between preexisting mood and media messages refers to the extent to which media messages "feature behaviors of great affinity to the mood state during

which they are received” (p. 154). When there is high semantic affinity between prior events leading up to the preexisting mood and media messages, exposure to messages is likely to contribute to rumination on mood-related thoughts and is thus less likely to alter the preexisting mood. Accordingly, media messages with low semantic affinity with previous experience are more capable of disrupting the preexisting mood. As such, high semantic-affinity content would be undesirable for individuals in a negative mood but ideal for individuals in a positive mood. MMT thus predicts that individuals in negative moods tend to be drawn to media content with minimal semantic affinity with the events leading to the negative mood, while those in positive moods tend to be drawn to content with high semantic affinity to the events inducing their positive mood.

Regarding the hedonic valence dimension, which refers to the positiveness/negativeness of media content, MMT posits that positively valenced (versus negatively valenced) media messages are more effective in disrupting preexisting bad moods and maintaining good moods. Therefore, individuals in negative affective states would likely be more drawn to positive content over negative content, whereas individuals in positive moods are less likely to display this preference (Zillmann 1988a).

Empirical research applying MMT has been growing over the past 30 years, with primary focus on examining the impact of mood on selective exposure to and engagement with entertainment media, and provides reasonably consistent evidence supporting significant impact of mood on selective exposure to different types of media content, primarily in the television context, but also including music, news, and the Internet (e.g., Knobloch and Zillmann 2002; Reinecke et al. 2012).

Applying the logic of MMT, this study predicts that, through the motivation to manage their affective states, consumers would likely be drawn to and engage with certain types of ads that are likely to enhance their moods. Specific hypotheses are posed for each mood-relevant characteristic dimension, predicting the impact of consumers’ temporary affective states on their selective attentive engagement with different types of ads. First, regarding the relationship between consumers’ affect and attention and engagement with ads with high versus low excitatory potential, research has indicated that, to maintain a pleasant state of excitement, individuals would be drawn to more exciting stimuli, while individuals in negative affective states would likely be drawn to more

soothing stimuli (e.g., Knobloch-Westerwick 2006). Thus, this study hypothesizes:

H1: Consumers in the positive affective state will have greater tendency to be drawn to engage with more exciting ads than those in the negative affective state.

Second, the mood-relevant attribute of ads related to absorption potential is the use of emotional versus rational appeals (e.g., Xie et al. 2004). Emotional appeals in advertising typically intend to stimulate consumers’ emotional responses (e.g., a cell phone ad dramatizes a person’s happy moments when using the phone), whereas rational appeals focus on providing factual information or compelling arguments regarding product attributes or utilitarian benefits of product usage (e.g., a cell phone ad emphasizes the phone’s superior functional features) (Moore, Harris, and Chen 1995; Xie et al. 2004). Compared to rational appeals, emotional appeals are more absorbing and engaging and would have higher absorption potential to disrupt and alleviate bad moods (Lee and Hong 2016). Because prior literature indicates that individuals in a negative mood would be drawn to absorbing stimuli that could disrupt the rehearsal of thoughts related to the original mood, while those in a positive mood would be more likely drawn to nonabsorbing stimuli (e.g., Knobloch-Westerwick 2006), the following hypothesis is offered:

H2: Consumers in the negative affective state will have greater tendency to be drawn to engage with ads using emotional appeals than those in the positive affective state.

Third, semantic affinity between an ad and a preexisting mood-inducing event is another mood-relevant attribute of ads. An example would be a connection between a mood induced by a favorite team being in the lead in during the Super Bowl and a Super Bowl-themed ad in the following commercial break. When semantic affinity is high, it would likely reinforce rehearsal of mood-related thoughts, which could noxiously contribute to a negative mood yet be more desirable for a positive mood. Consequently, individuals in a negative mood tend to disrupt the existing mood by selecting media content with minimal semantic affinity with the events inducing the negative mood, while those in a positive mood might seek to maintain prior mood by seeking media content with high semantic affinity (e.g., Knobloch-Westerwick 2006). Therefore, it is predicted:

H3: Consumers in the positive affective state will likely be drawn to engage with ads with high semantic affinity with the prior event generating the affective state, whereas those in the negative affective

state will likely be drawn to engage with ads with lower semantic affinity.

Fourth, ads can be categorized into positive and negative valence. While pleasant ads can have an uplifting effect and raise consumers' spirits, ads laden with negative emotions such as sadness and anxiety could bring consumers' emotions down. Research has demonstrated that individuals in a negative mood will seek to disrupt it by exposure to more positively valenced stimuli, whereas those in a positive mood might exhibit less of such tendency (e.g., Knobloch and Zillmann 2002). For example, in an experiment conducted by Knobloch and Zillmann (2002), participants in the bad mood condition spent more time listening to high-arousal, positively valenced music (i.e., energetic-joyful music) over music low in these qualities to a greater degree than did participants in the good mood condition. A recent eye-tracking study also showed that participants induced into negative affect tended to pay more attention to happy faces, suggesting the joint influence of participants' affect and the emotional valence of the visual stimuli on attention (Sanchez et al. 2014). Thus, this study hypothesizes:

H4: Consumers in the negative affective state will have greater tendency to be drawn to engage with positively valenced ads than those in the positive affective state.

Method

Most of the previous studies on affective factors in advertising have used experimental methods where affective states are either manipulated or measured by self-report measurements, which rely on participants' cognition (Knobloch-Westerwick 2006). While such methods have advantages in internal validity threat control, major disadvantages are external and ecological validity issues and inability to measure and test actual, real-time temporary affective fluctuation and its real-time impact on consumers' reactions to ads.

To overcome the methodological limitations of the traditional experimental method and to advance theory building on the influence of affect on consumers' reactions to ads, this study devised an innovative computational research approach to examine the effects of consumers' real-time temporary affective states on relative attentional engagement with different types of ads in a natural setting, by analyzing TV content data (both the TV program and accompanying ads) and Twitter data of consumers' tweets during the broadcast of the 2019 Super Bowl.

The reason for choosing the Super Bowl broadcast as the study's context is twofold: (1) in such a high-stakes sporting event, consumers' affective states can fluctuate frequently and extensively depending on the whether their affiliated teams are winning or losing throughout, which offers a great proxy measure for real-time affective states; and (2) given that high-stakes sports events are broadcast live and watched in real time across the country (and even around the world), mass audiences are exposed to the same set of nationally broadcast commercials, which provides a perfect opportunity for matching proxy measures of real-time affective fluctuation and related ad responses captured from massive social media data. Furthermore, the Super Bowl broadcast is an advertising event as much as a sports event, and ads placed during the Super Bowl are simultaneously viewed by a massively large audience and tend to spark a large volume of social media conversations in a short period of time.

Data Collection

Three types of data were collected to compute proxy measures for the independent variable (consumers' real-time affective states) and dependent variable (real-time relative engagement with different types of ads) and to determine mood-relevant characteristics of the ads aired during the Super Bowl to classify ad types. First, the airtime schedule of the broadcast, including both the full Super Bowl game and all of the commercial breaks, was obtained. Second, TV content data (both the game and accompanying ads) were obtained. The Super Bowl game itself was analyzed for determining each team's performance at the moment of each commercial break, and the ads were analyzed for determining mood-relevant characteristics and orienting-attention-grabbing features.

Third, viewers' Twitter data reacting to the Super Bowl game and commercials were obtained using Twitter streaming API via the AIDR tool (Imran et al. 2014). The set of keywords and hashtag search strings used to collect all relevant tweets from the start of the broadcast (February 2nd, 5:30 p.m. CST) to the end of the day was decided in two ways (refer to [Supplemental Online Appendix](#)): First, an initial list was determined based on pregame news coverage and typical trending Super Bowl hashtags, such as #SuperBowl; second, brands and prominent copy and images used in the commercials were collected in real time and added to the search terms on AIDR. Given that consumers often comment on specific features (e.g., songs, singers, celebrities, images, or specific

scenes) in ads without mentioning the advertised brands, we developed a secondary ad keyword set consisting of words and phrases relevant to the commercials. This keyword development approach enabled a more inclusive and thorough collection of ad-related tweets than the approach solely relying on brand names.

Content Analysis of Mood-Relevant Characteristics of Ads

All of the national commercials that aired during the Super Bowl broadcast were content-analyzed on the four dimensions of mood-relevant characteristics of ads: excitatory potential, use of emotional (versus rational) appeal, semantic affinity, and hedonic valence. The coding scheme and detailed coding instructions were developed based on the MMT research literature (e.g., Bryant and Zillmann 1984). Two undergraduates were hired and trained to serve as independent coders. After several rounds of training sessions, the coders coded all of the commercials.

Excitatory potential of ads was coded using eight items (e.g., relaxing, stimulating, and arousing) on a 7-point scale ranging from (1) *Not at all* to (7) *Extremely* (Bryant and Zillmann 1984). Interrater reliability was tested using Krippendorff's alpha, yielding seven acceptable scores: 0.94 (relaxing), 0.77 (stimulating), 0.82 (arousing), 0.93 (peaceful), 0.85 (exciting), 0.91 (unwinding), and 0.89 (restful). The score for emotionally involving was 0.54; thus, this item was excluded from data analysis. Then, upon confirming acceptable measurement reliability (Cronbach's $\alpha = .96$), a summated *excitatory potential* score was computed by averaging the seven item scores ($M = 4.84$, $SD = 1.34$). For hypothesis testing, this summated score was used to classify each ad into the exciting or soothing ad category by using the scale middle point of 4 as the split point.

Use of emotional (versus rational) appeal was coded using a 7-point scale, where 1 represents *Purely emotional*, 4 represents *A balance of emotional and rational*, and 7 represents *Purely rational* (Lang 1990). The interrater reliability score (Krippendorff's alpha = 0.93) was acceptable. For ad classification, this variable was later recoded into a binary *Emotional-Rational ad* variable using the scale middle point of 4 as the splitting point.

For *semantic affinity*, each ad was classified into one of three categories: football-themed ads, sports-themed ads (other than football), and non-sports-themed ads (Krippendorff's alpha = 0.90). This variable was then recoded into a binary *high versus low semantic-affinity ad* variable, with football-themed and sports-themed ads combined into high semantic

affinity and non-sports-themed ads classified as low semantic affinity. For *hedonic valence*, each ad was coded into negative, neutral, positive, or mixed based on the overall tone of the ad (Krippendorff's alpha = 0.89). Similarly, this variable was collapsed into a binary *Positive/Nonpositive ad* variable: commercials with positive valence were categorized as positively valenced ads, while commercials with negative, neutral, or mixed values were classified as non-positively valenced ads.

In addition, the presence/absence of certain orienting cues in the commercials was coded to generate control variables to be entered in the hypothesis testing, as prior research identified some significant influence of orienting-eliciting features in media content (i.e., specific media messages/stimulus feature) on viewers' attention (e.g., Lang et al. 1999; Thorson and Lang 1992). Specifically, the following ad content features were coded: use of music (fast paced, slow paced or no music), use of loud sound, production pacing (fast, medium, or slow), use of a celebrity, and use of novel elements.

Data Analysis and Results

Proxy Measure Computation

Independent Variable: Real-Time Affective State

Affective fluctuation often occurs while viewing a high-stakes sports event such as the Super Bowl, where viewers' (especially those emotionally involved and rooting for a particular team) temporary affective states are linked to the moment-by-moment performance of their affiliated teams as the game progresses. Thus, affective states of viewers who tweet during the broadcast can be inferred based on their team affiliation in conjunction with the real-time performance of competing teams.

Therefore, the first step in computing the affective state proxy measure is to determine each viewer's team affiliation (i.e., rooting for the New England Patriots or the Los Angeles Rams), which was done by analyzing Twitter follower list data: following one of the two competing teams' official Twitter accounts (@Patriots or @RamsNFL), its mascot's official Twitter account (@PatPatriot or @RampageNFL), or verified Twitter accounts of the teams' players. To deal with cases where multiple teams' accounts are followed, we developed the following rules for deciding team affiliation:

- When a user followed multiple accounts associated with a single team, the user was considered a fan

of the team with which the accounts were associated (Patriots or Rams).

- When a user followed multiple accounts associated with both teams and followed a key player from one team (e.g., the Patriots), the user was considered more of a Patriots fan than a Rams fan, because this was considered to indicate a higher degree of investment in that team and therefore a stronger affiliation to that team.
- When a user followed multiple accounts associated with both teams and also followed key players from both teams, the user was considered a fan of both teams and thus not included in either team fan group.
- When a user followed multiple accounts associated with both teams and followed no key players, the user was considered a general football fan and thus not included in either team fan group.

Once team affiliation was determined, the second step was to match it to the real-time performance of the competing teams during the game's progression and determine each team fan group's estimated affective state around each commercial break. Specifically, when there was a score change right before a commercial break, Twitter users whose following data were indicative of rooting for the team that just scored would be considered to be in a positive affective state, while Twitter users whose following data were indicative of rooting for the competing team would be considered to be in a negative affective state. Based on such a matching procedure, each fan group's changing affective states were coded throughout the game. For commercial breaks immediately following a play with no scoring, affective states were coded based on the score as of when the commercial break came on. Specifically, if one team was in the lead when a commercial break came on, Twitter users whose following data were indicative of rooting for that team were considered in a positive affective state. Similarly, when the game was tied when a commercial break came on, a neutral affective state was assigned to supporters of both teams. Thus, for each commercial break, this variable was coded as (1) positive, (2) neutral, or (3) negative affective state. Following this approach, the real-time affective fluctuation data were computed for the entire duration of the broadcast.

Dependent Variable: Selective Engagement with Different Types of Ads

In the computational research approach, relying on digital footprint data left by consumers' online

behaviors, researchers have been developing new methods of data analysis for the purpose of expanding existing theories and making new data-driven discoveries (Humphreys and Wang 2018). Utilizing online behavioral data for theory testing entails mapping a theoretical construct to available and appropriate proxy measures, as it is impossible to find a direct measure indicating abstract concepts like engagement (Huh 2017).

The common proxy measures used by marketing, advertising, HCI, and media studies researchers for consumers or media users' engagement with online ads or digital media content have been clicking behaviors and social media liking, commenting, or sharing behaviors (e.g., Hwang et al. 2017; Ksiazek, Peer, and Lessard 2016), with the latter three being commonly used in the behavioral analytics metrics to measure social media marketing success. Following this approach, a proxy measure for relative selective engagement with different types of ads according to mood-relevant characteristics was developed by analyzing each team-affiliated consumer's tweets regarding the ads aired during each commercial break. For example, given five different ads airing during a commercial break, with two being exciting ads and three being nonexciting ads, if one group of viewers in a positive affective state posted relatively more tweets about the two exciting ads than about the three nonexciting ads, at the aggregate level we could take such data as indicating relatively higher engagement with exciting ads than with nonexciting ads for this group.

Developing this dependent variable proxy measure entailed the following computational data analysis steps: (1) filtering ad-related tweets that are correctly matched to nationally aired commercials during the Super Bowl; (2) mapping each of the ad-related tweets to the right corresponding ad; and (3) classifying the ad-related tweets into different ad types according to each of the four mood-relevant characteristic dimensions.

The automatic filtering of ad-related tweets and tweet-to-ad mapping required first developing a comprehensive list of ad-related keywords that capture the unique characteristics of each ad, such as celebrities appearing in the ads, prominent images or copy, and cultural symbols like classical songs. Several rounds of pretests of the ad keyword set were conducted with subsamples of the data set, where manual identification of key descriptive words of the ads was performed. The final ad keyword set was also used later as an input for the deep-learning approach to automatically map each tweet to its corresponding ad. See

the [Supplemental Online Appendix](#) for a figure illustrating the detailed steps involved in the computational approach.

First, all of the collected tweets were subjected to automatic filtering through natural language processing (NLP) steps to retain only ad-related tweets. The first step was removal of retweets and duplicate tweets, followed by a standard preprocessing stage, as tweets tend to include symbols and artifacts that can confuse the classification process. We stripped the lowercase tweets of username handles and punctuation (including emojis), URLs, words smaller than two letters, words that contain numbers, and non-ASCII characters, followed by removal of English stop words as well as data-set-specific stop words. Next, to narrow down on possibly ad-related candidate tweets we developed a set of heuristics (e.g., checking for presence of brand names). Then, we checked for the presence of a high degree of overlap between the ad keyword set and each tweet using the Jaccard Index measure (Niwattanakul et al. 2013).

After the ad-related tweet filtering process, the ad-related tweets were subjected to the process of tweet-ad mapping through a deep-learning approach. To create a training data set for the deep-learning approach, a random sample of the tweets was drawn for manual labeling by three independent coders such that each tweet was assigned to the ad it referred to, or labeled as “none” if the tweet was not ad related. This manual labeling was done based on the ad keyword set as well as the coders’ additional notes prepared on viewing the commercials. Any differences in labeling were settled by a majority vote among the coders to ensure reliability. Detailed numbers about the training data are presented in the [Supplemental Online Appendix](#). Finally, these manually labeled data were divided into training and test sets and used by the deep-learning approach, which is explained next.

Deep-Learning Model to Classify Tweets and Map to Corresponding Ads. To classify and map each ad-related tweet to the corresponding ad, we used Google’s BERT (Bidirectional Encoder Representations from Transformers; Devlin et al. 2018), one of the most popular deep-learning models that follows a bidirectional approach to read the entire sequence of words in the text at once, and outperforms other traditional NLP models in terms of taking into consideration the contextual meaning of a word in a sentence.

Learning the contextual meaning of words involves training the NLP model on all possible usages of words in a sentence, which is a very computationally

intensive task. BERT is a large neural network with a huge number of parameters, ranging from 100 million to over 300 million. Thus, training a BERT model from scratch would result in overfitting. Instead, we decided to fine-tune (Sun et al. 2019) the pretrained BERT-base-uncased model (i.e., trained on a huge data set; Devlin et al. 2018) with 12 layers on our relatively smaller data set and avoided overfitting by considering larger batches. In this iteration of the analysis, we used a modified training strategy with fine-tuned BERT, because our use case involved mapping tweets to 61 ads that were commented upon, which is a relatively large number of classes. One thing worth noting is that some of the ads were frequently discussed on Twitter while others were seldom commented upon, resulting in an unbalanced representation for different ads. To address this, we modified the training strategy in the way we used BERT.

To provide a better training signal from the limited labeled data set we had, we proposed learning from the textual descriptions of each ad class, which is a part of our ad keyword set. We framed the multiclass classification task in terms of a binary classification task: Specifically, the input is a *<tweet, ad-description>* pair, and the output is either **True** (if the tweet is related to the ad in the included ad description) or **False** (otherwise). If the tweet is not related to any ad, it is categorized as belonging to the “none” class; otherwise, the tweet is classified as “ad related.” Therefore, given N tweets and K ad categories, the BERT model would be trained on NK instances and the training sample is therefore exploded. This training strategy makes no assumptions on the number of ad categories and is easily extensible. For example, including new ad categories or adapting to newer ad themes would require only a modification in the ad descriptions with little to no fine-tuning of the classifier itself. This approach automatically mapped each tweet to its corresponding ad. A detailed description of the performance of the deep-learning model on ad classification of tweets is included in the [Supplemental Online Appendix](#).

Descriptive Statistics from the Ad Content Analysis

There were 95 nationally aired commercials in total that were placed in 27 commercial breaks. The length of the commercials ranged from 5 to 120 seconds, with the majority of commercials lasting for 30 seconds. The content analysis of the Super Bowl ads showed some interesting patterns. As for the ads’

Table 1. Descriptive statistics of Twitter data.

	New England Patriots	Los Angeles Rams	Total
Number of Super Bowl-related tweets identified to be affiliated with one of the competing teams	22,351	5,855	28,206
Number of ad-related tweets	3,653	983	4,636
Number of Twitter accounts associated with Super Bowl-related tweets	18,706	4,897	23,603
Number of Twitter accounts associated with ad-related tweets	3,321	891	4,212

Note: Time frame for the collected tweets is 5:30 p.m. CST to 12:00 a.m. CST on Super Bowl Sunday.

excitatory potential, they tended to be more exciting (61.1%) than soothing. Between the emotional and rational appeals, a large majority of the ads used emotional appeals (75.8%). As for semantic affinity with the game, while 85.3% of the ads were not related to any sports themes, some commercials used sports-related themes (either football or sports in general) to appeal to game viewers (14.7%). About three-quarters of the ads used positively valenced tones (74.7%).

Descriptive Statistics of the Twitter Data

A total of 972,003 Super Bowl-related tweets were collected during the time frame, of which 47,660 tweets were identified using the follower list-based team affiliation measure. Among those tweets, 28,206 tweets were identified to be affiliated with either of the competing teams. From the Super Bowl-related tweets, a subset of ad-related tweets was generated using our ad classification methods described previously. Descriptive statistics of the Twitter data are presented in Table 1.

Hypothesis Testing Results

Hypothesis 1: Consumers' Affect and Engagement with Exciting versus Soothing Ads

Hypothesis 1 predicted that consumers in the positive affective state would have greater tendency to be drawn to engage with more exciting ads than those in the negative affective state. To test the hypothesis, we first visualized the overall pattern of aggregate data indicating each fan group's engagement with different types of ads within each commercial break (see the [Supplemental Online Appendix](#)), with the *x*-axis representing the commercial breaks that contained both exciting and nonexciting ads, the *y*-axis representing the percentage of mentions about exciting ads for each group in each commercial break, and the overlay of each group's fluctuating affective states illustrated using colored dot marker based on the real-time performance of the two competing teams (the green dot marker indicates positive affect; gray is neutral state; and red is negative affect).

Among the 17 commercial breaks that contained both exciting and nonexciting ads, in 12 commercial breaks the two team groups were in different affective states and were included in the analysis. A careful examination of the fluctuation of the distribution of each group's percentages of mentions about exciting ads reveals that the positive affective group had higher percentage of exciting ad mentions than did the negative affective group in five of the 12 commercial breaks, while the negative affective group had higher percentage of exciting ad mentions than did the positive affective group in another five commercial breaks. In the remaining two commercial breaks, both groups had the same percentage of exciting ad mentions. The descriptive data do not show any consistent pattern supporting hypothesis 1.

Next, a hierarchical logistic regression analysis was performed with engagement with exciting versus soothing ads as the dependent variable, all the five control variables entered in the first block and the independent variable of affective state entered in the second block. The overall model significantly predicted engagement with exciting (versus soothing) ads ($\chi^2(7) = 785.30, p < .001$, Nagelkerke's $R^2 = 0.56$). As presented in Table 2, however, while the control variables of use of celebrities, use of music, inclusion of novel elements, and loud sound were significantly related to engagement with exciting versus soothing ads, the independent variable, affective state, was not. Therefore, hypothesis 1 was not supported.

Hypothesis 2: Consumers' Affect and Engagement with Emotional versus Rational Ads

Hypothesis 2 predicted that consumers in the negative affective state would have greater tendency to be drawn to engage with ads using emotional appeals than those in the positive affective state. First, the distribution of each group's percentages of mentions of emotional ads was analyzed across 18 commercial breaks that contained both types of ads, among which the two groups differed in affect in 12 commercial breaks. The overall aggregate data pattern showed that the negative affective group had a higher percentage of emotional ad mentions than did the positive

Table 2. Hierarchical logistic regression for affective state and engagement with exciting versus soothing ads.

Predictors	Model 1			Model 2		
	B	SE	Wald	B	SE	Wald
Use of celebrity	2.28***	.19	138.02	2.28***	.19	137.86
Inclusion of novel elements	.82***	.22	13.46	.84***	.22	13.95
Use of music			174.74***			175.07
Use of music (1)	−20.22	2618.41	.00	−20.24***	2613.41	.00
Use of music (2)	−17.77	2618.41	.00	−17.76	2613.41	.00
Loud sound	−19.35	2618.41	.00	−19.35	2613.41	.00
Production pacing	.35	.25	1.84	.35	.26	1.89
Affective state				−.22	.19	1.41
Cox and Snell R^2		.36			.36	
Model statistics	−2LL = 1045.72; $\chi^2 = 783.90$; $df = 6$; $p < .001$			−2LL = 1044.32; $\chi^2 = 785.30$; $df = 7$; $p < .001$		

Note: Use of music consists of three values and was dummy-coded; *** $p < .001$.

affective group in only four of the 12 commercial breaks (see the [Supplemental Online Appendix](#)). However, in seven commercial breaks, the positive affective group had a higher percentage of emotional ad mentions than did the negative affective group. In the remaining one commercial break, the two groups had the same percentage of emotional ad mentions. This pattern was inconsistent with hypothesis 2.

To further test the hypothesis, a hierarchical logistic regression analysis was performed, using the same set of control variables and approach used in testing hypothesis 1. The overall model significantly predicted engagement with emotional (versus rational ads) ($\chi^2(7) = 684.25$, $p < .001$, Nagelkerke's $R^2 = 0.46$). As presented in [Table 3](#), while the control variables of use of celebrity, use of music, inclusion of novel elements, and production pacing were significantly related to engagement with emotional versus rational ads, the independent variable of affective state was not. Therefore, hypothesis 2 was not supported.

Hypothesis 3: Consumers' Affect and Engagement with High versus Low Semantic-Affinity Ads

Hypothesis 3 predicted that consumers in the positive affective state would likely be drawn to engage with high semantic-affinity ads, whereas those in the negative affective state would be drawn to low semantic-affinity ads. The distribution of each group's percentages of mentions of high semantic-affinity ads across the 11 commercial breaks that contained both types of ads reveals (see the [Supplemental Online Appendix](#)) (1) the positive affective group had higher percentages of high semantic-affinity ad mentions than did the negative affective group in four out of the eight commercial breaks where different affective groups could be compared; (2) for two commercial breaks, the positive affective group had lower percentage of high semantic-affinity ad mentions than did the negative affective group; and (3) in two remaining commercial breaks both groups did not mention any high semantic-affinity ads. This

distribution seems to show a pattern that is reasonably consistent with hypothesis 3.

The hypothesis was further tested with the same hierarchical logistic regression analysis as noted previously. The overall model significantly predicted engagement with high semantic-affinity ads (versus low semantic-affinity ads) ($\chi^2(7) = 1175.72$, $p < .001$, Nagelkerke's $R^2 = 0.69$) (see [Table 4](#)). All of the control variables were significantly related to engagement with high versus low semantic-affinity ads, but even after these variables were controlled for, affective state emerged as a significant predictor ($B = .40$, Wald = 4.27, $p < .05$, Exp(B) = 1.49). The results revealed that consumers in a positive affective state would more likely be drawn to engage with high semantic-affinity ads compared to those in a negative affective state. Therefore, hypothesis 3 was supported.

Hypothesis 4: Consumers' Affect and Engagement with Positive versus Negative Ads

Hypothesis 4 predicted that consumers in the negative affective state would have greater tendency to be drawn to engage with positively valenced ads than those in the positive affective state. The distribution of each group's percentages of mentions of positively valenced ads was analyzed across 16 commercial breaks that contained both types of ads, among which the two groups had different affective states in 10 commercial breaks. Examination of the distribution of each team's percentages of mentions of positively valenced ads revealed that the negative affective group mentioned a higher percentage of positively valenced ads than did the positive affective group in six of the 10 commercial breaks (see the [Supplemental Online Appendix](#)). However, in one commercial break, the positive affective group mentioned a higher percentage of positively valenced ads than did the negative affective group; in the remaining three commercial breaks, both groups had the same percentage of positively

Table 3. Hierarchical logistic regression for affective state and engagement with emotional versus rational ads.

Predictors	Model 1			Model 2		
	B	SE	Wald	B	SE	Wald
Use of celebrity	.58***	.15	14.15	.59***	.15	14.73
Inclusion of novel elements	−1.60***	.17	84.27	−1.61***	.17	85.50
Use of music			20.184***			21.22***
Use of music (1)	−19.32	2367.77	.00	−19.32	2366.25	.00
Use of music (2)	−20.02	2367.77	.00	−20.04	2366.25	.00
Loud sound	−23.68	2367.77	.00	−23.72	2366.25	.00
Production pacing	4.16***	.32	166.43	4.16***	.32	166.56
Affective state				.239	.155	2.38
Cox and Snell R^2		.33			.33	
Model statistics	−2LL = 1432.60; $\chi^2 = 681.84$; $df = 6$; $p < .001$			−2LL = 1430.19; $\chi^2 = 684.25$; $df = 7$; $p < .001$		

Note: Use of music consists of three values and was dummy-coded; *** $p < .001$.

Table 4. Hierarchical logistic regression for affective state and engagement with high versus low semantic-affinity ads.

Predictors	Model 1			Model 2		
	B	SE	Wald	B	SE	Wald
Use of celebrity	1.49***	.23	42.56	1.50***	.23	43.12
Inclusion of novel elements	−2.60***	.20	177.99	−2.63***	.20	178.61
Use of music			17.40***			17.20***
Use of music (1)	−19.15	5151.61	.00	−19.18	5139.79	.00
Use of music (2)	−14.92	5151.61	.00	−14.98	5139.79	.00
Loud sound	.76**	.28	7.34	.73**	.28	6.61
Production pacing	4.00***	1.03	15.01	3.99***	1.03	14.98
Affective state				.40**	.19	4.27
Cox and Snell R^2		.49			.49	
Model statistics	−2LL = 993.65; $\chi^2 = 1171.36$; $df = 6$; $p < .001$			−2LL = 989.29; $\chi^2 = 1175.72$; $df = 7$; $p < .0011$		

Note: Use of music consists of three values and was dummy-coded; *** $p < .001$; ** $p < .05$.

valenced ads. The pattern seems to be consistent with hypothesis 4.

To further test the hypothesis, a hierarchical logistic regression analysis was performed. The overall model significantly predicted engagement with positively valenced (versus non-positively valenced) ads ($\chi^2(7) = 906.84$, $p < .001$, Nagelkerke's $R^2 = 0.65$). As Table 5 shows, among the control variables, use of music, inclusion of novel elements, and loud sound were significantly related to engagement with positively valenced versus non-positively valenced ads. After controlling for these, affective state was still a significant predictor ($B = -.71$, Wald = 10.14, $p < .05$, Exp(B) = .493). The results revealed that consumers in a negative affective state would more likely be drawn to engage with positively valenced ads compared to those in a positive affective state. Therefore, hypothesis 4 was supported.

Discussion

Facing growing challenges in getting consumers' engaging reactions in the increasingly cluttered media environment, consumer attention and reactions to ads (or lack thereof) and influencing factors have attracted a great deal of attention by advertising scholars and professionals. While important strides have been

made in the literature to better understand influencing factors of consumers' ad reactions, one of the most underresearched, yet critically important, factors is consumers' temporary affective state during ad exposure as a potentially important context factor. To fill this gap in the literature this study examined the impact of consumers' temporary affective states experienced during a TV watching situation on relative engagement with different types of ads with different mood-relevant characteristics, using an innovative computational research approach.

The findings shed empirical light on the underexplored area of research on affect and ad engagement, and reveal important insights. First, even when exposed to the same set of ads, consumers in the positive affective state showed greater tendency to be drawn to engage with ads with high semantic affinity (versus low semantic-affinity ads) than those in the negative affective state. This finding is consistent with previous MMT studies examining the effects of affective states on selective exposure choices to media content (e.g., Knobloch-Westerwick 2006). In addition, consumers in the negative affective state had greater tendency to be drawn to positively valenced ads (versus non-positively valenced ads) than those in the positive affective state, which is in line with the MMT literature suggesting that individuals in negative

Table 5. Hierarchical logistic regression for affective state and engagement with positive versus nonpositive ads.

Predictors	Model 1			Model 2		
	B	SE	Wald	B	SE	Wald
Use of celebrity	-.192	.20	.97	-.25	.20	1.59
Inclusion of novel elements	-1.19***	.26	20.25	-1.17***	.27	19.52
Use of music			267.95***			264.69***
Use of music (1)	20.04	5161.01	.00	20.12	5127.12	.00
Use of music (2)	23.93	5161.01	.00	24.15	5127.12	.00
Loud sound	1.40***	.28	25.04	1.49**	.28	27.59
Production pacing	-20.61	2297.36	.00	-20.67	2277.63	.00
Affective state				-.71**	.22	10.14
Cox and Snell R^2		.40			.41	
Model statistics	-2LL = 820.71; $\chi^2 = 896.54$; $df = 6$; $p < .001$			-2LL = 810.41; $\chi^2 = 906.84$; $df = 7$; $p < .001$		

Note: Use of music consists of three values and was dummy-coded; *** $p < .001$; ** $p < .05$.

moods are more drawn to positively valenced media content to improve their current mood state (e.g., Biswas, Riffe, and Zillmann 1994; Knobloch and Zillmann 2002; Zillmann, Hezel, and Medoff 1980).

However, the other two hypotheses were not supported. Particularly, the findings did not support the predicted influence of consumers' affective states on selective engagement with exciting versus soothing ads or emotional versus rational ads. The null finding on the emotional versus rational dimension of ads could be explained by the possibility that the dimension of absorption potential might be more subtle and complex than other mood-impacting characteristic dimensions, resulting in a considerable complication when testing hypotheses related to it. This is in line with complexities about the absorption potential dimension reported in prior MMT research, which suggests that the complicated nature of media content precludes the isolation of some dimensions, particularly the absorption potential dimension, from others such as excitatory potential and valence dimensions (Knobloch-Westerwick 2006). If this is indeed the case, detection of a significant effect of affective state on attentional engagement with emotional versus rational content might be possible only with controlled experiments where the other mood-impacting dimensions are held constant. The lack of supporting evidence for the exciting versus soothing dimension of ads might be attributable to the unique viewing situation of Super Bowl games where people typically watch together with friends and families and might be generally more attracted to more exciting commercials.

Theoretical Implications

This study offers several important theoretical implications to the advertising effect literature and application of the MMT to advertising research. First, this study contributes to advancing the ad engagement research by examining the largely untested affective side of influencing factors and demonstrating the

linkage between affect and selective engagement with different types of ads along the semantic affinity and valence dimensions of mood-relevant characteristics. Despite growing research on influencing factors of consumers' ad attention and engagement, research on the influence of consumers' affective factors is relatively more limited than research on the cognitive factors, with the role of consumers' affective states at the point of ad exposure being especially underexplored. Given that feelings permeate consumers' daily life and consumers are likely to be exposed to ads in the presence of some types of affect, examining the impact of consumers' temporary affect on their selective attention to and engagement with ads provides another important window into understanding consumers' responses to different types of ads.

Second, this study extends the research on mood management effects to the advertising context. While empirical evidence on the linkage between affective states and selective exposure to media content from MMT application works (e.g., Knobloch-Westerwick 2006) and empirical evidence connecting affective states and attention to visual stimuli (e.g., Sanchez et al. 2014) is accumulating, little research has explored this in the context of advertising. Advertising, as a type of media content, might have more mood-relevant characteristics that are likely to elicit different attentional processing patterns than emotional valence alone. This study specifies four theory-derived characteristics of ads that are likely to facilitate viewers' mood management goals, two of which—semantic affinity and valence—received empirical support. This is an important finding as it shows that selective engagement with ads was also linked to mood management motivation, despite their persuasive nature and being less likely to be actively sought out by consumers.

Third, this study offers a unique methodological contribution to advancing the research on advertising effects and influence of affect by developing a novel computational research approach that allows for the

matching of real-time TV watching and inferred affective state fluctuation, and ad engagement. Specifically, (1) our innovative state-of-the-art deep-learning algorithm for tweet-ad matching overcomes the commonly used brand name-based ad annotation approach and makes a significant contribution to advancing the analysis of social media data to monitor and detect consumers' reactions to ads; (2) our affective state proxy measure by matching media viewers' estimated emotional involvement (being a fan of a competing team) and the media content context (the real-time performance of the team) is a novel methodological innovation that can be applied to future computational research and context-based computational advertising. Following the theory-guided concept-proxy mapping process, this study suggests a new promising avenue of research to apply the computational research approach to advertising theory building, especially theory regarding affective factors influencing consumers' reactions to ads.

Practical Implications

Practically, the results of this study provide significant implications for ad targeting and placement strategies based on consumers' temporary affective states induced by media consumption. This offers important implications for future computational advertising model development, especially in connection to emerging computer science research on context-aware recommender systems and affect-based ad personalization. This study's findings propose a new promising way to target consumers based on their current affective states that can be captured by appropriate proxy measure data that could be potentially obtained from various consumer touchpoints. With the availability of real-time consumer media use data, the rapid development of data-driven targeting strategies, and increasing use of big data for advertising, extremely personalized, real-time media content consumption-related targeting will be more frequently used than ever.

Limitations and Suggestions for Future Research

While it is meaningful that this study took the first step toward applying a computational research approach to examining consumers' real-time affective fluctuation and connecting it to real-time ad engagement, it can be improved in a number of ways. First, because the findings of this study are based on a single sports event broadcast, the findings might be unique to this type of sports event broadcast. Moreover, the fact that the Super Bowl broadcast was

chosen due to its large viewership and reach also means that this study's results might not be directly generalizable to other types of TV viewing situations. This study also acknowledges the limitation that not everyone who watched the Super Bowl would engage in social media posting activities. In other words, Twitter data do not fully represent the Super Bowl viewer population.

Second, while this study used a state-of-the-art deep-learning model for ad-tweet mapping, one constraint was that we did not consider tweets mentioning multiple ads as training data to avoid confusion in the model's predictions. There might be tweets mentioning more than one ad, and the model would need to be appropriately modified to handle such cases in the future. Another related limitation is that, while we carefully developed our proxy measures for affective state and ad engagement, the measurements suffer from weaknesses inherent in any computational proxy measures and have room for improvement. Appropriate measurement validity testing needs to be conducted by comparing our proxy measures against other measurement approaches.

Third, while the computational research approach adopted in this study has certain advantages in terms of ecological and external validity, it lacks control on confounding factors, such as age. Because prior research showed that older adults were better at regulating their negative affective states (Demeyer, Sanchez, and De Raedt 2017), age might be an important factor to consider. Future studies are encouraged to develop methodological triangulation to test the results of this study by directly measuring and testing the influence of temporary affective state on ad reactions in a natural setting, such as during actual Super Bowl game viewing.

Despite the aforementioned methodological limitations, the current study provides valuable insight and unique empirical evidence supporting the impact of consumers' temporary affective states during ad exposure on their selective ad reactions. More research is needed to uncover the influence of consumers' incidental affect on their reactions to ads and offer practical implications for affect-based ad targeting strategy.

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